HO SCALE RAILROAD — MASTER CONTROLLER & WIRING

Raspberry Pi 4B • 12 V DC track • 5 V motor/servo system • A–G powered turnouts • rotary tipper • yard power

POWER DISTRIBUTION

12 V / 18 A TRACK PSU
- MAIN branch: 7 A fast-blow -> main feeder bus
- TIPPER branch: 2 A fast-blow -> isolated tipper rails
- 1/6 in air gap at BOTH ends of tipper rail section

5 V / 18 A AUX PSU
- Pi 4B branch
- Servo PCA 0x42: 5 A fused branch
- DRV8833 #1–#8: 1 A fast-blow EACH
- PCA/MCP logic branches

16 V DECOUPLER SUPPLY
- Internal fuse -> relay contacts -> decoupler coil

GROUND
- Common star-ground bus for Pi / 5 V / 12 V / 16 V control returns

RASPBERRY PI 4B

I²C: GPIO2 SDA / GPIO3 SCL
GPIO22 Tipper IR Front
GPIO23 Tipper IR Back
GPIO24 Tipper Hall Back
GPIO25 Tipper Hall Front
GPIO5 Tipper Hall Clear
GPIO17 Tipper encoder
GPIO6 Decoupler relay
GPIO12 Yard relay 1 *
GPIO13 Yard relay 2 *
GPIO16 Yard relay 3 *

*** = newly assigned in this master pinout**

I²C BUS

Pi 3.3 V SDA/SCL -> bidirectional level shifter
-> 5 V I²C bus: 0x20, 0x21, 0x40, 0x41, 0x42
All boards share SDA, SCL and common ground

PCA9685 0x40 — MOTOR M1

CH0/1 Mine lift -> DRV8833 #1
CH2/3 Ski chairlift -> #2
CH4/5 Grain auger -> #3
CH6/7 Farm road -> #4
CH8/9 Ski lodge road -> #5
CH10/11 Long road -> #6
CH12/13 Town road -> #7
CH14/15 Rotary tipper -> #8

PCA9685 0x41 — MOTOR M2

Installed/reserved motor-expansion board
CH0–CH15 currently SPARE

PCA9685 0x42 — SERVOS

CH0 Farm gate CH4 Town gate
CH1 A CH2 B CH3 C
CH5 Yard1 CH6 Yard2 CH7 Yard3
CH8 Tipper wheel lock
CH9 D CH10 E CH11 F CH12 G
CH13–15 SPARE

MCP23017 0x20 — INPUTS 1

A: GPA0/1/2 = Common/Straight/Diverging
B: GPA3/4/5 = Common/Straight/Diverging
C: GPA6/GPA7/GPB0
D: GPB1/2/3
E: GPB4/5/6
GPB7 SPARE
Track Hall inputs: active-low

MCP23017 0x21 — INPUTS 2

F: GPA0/1/2
G: GPA3/4/5
GPA6 Mine Hall *
GPA7 Farm Hall *
GPB0 Ski Hall *
GPB1 Town Hall *
GPB2–GPB7 SPARE
Track Hall inputs: active-low

ROTARY TIPPER

Motor: DRV8833 #8 from PCA 0x40 CH14/15
Wheel-lock servo: PCA 0x42 CH8
IR Front -> GPIO22 (clear HIGH)
IR Back -> GPIO23 (clear HIGH)
Hall Back -> GPIO24 (active LOW)
Hall Front -> GPIO25 (active LOW)
Hall Clear -> GPIO5 (active LOW)
Encoder -> GPIO17
Decoupler relay -> GPIO6
Track section isolated by 1/6 in rail gaps
Dedicated 12 V / 2 A fused rail feed

YARD POWER

Exactly 3 switched DC yard tracks:
Track 1: Stall 1A + Stall 1B share relay
Track 2: Stall 2
Track 3: Stall 3

GPIO12 -> Yard relay 1 *
GPIO13 -> Yard relay 2 *
GPIO16 -> Yard relay 3 *
Relay contacts switch 12 V rail feed
Use relay driver / opto interface; do not drive coils directly from Pi

TURNOUTS / GATES

Powered turnout servos: A–G
Passive splitters: 3; NO wiring

0x42 CH1–3 = A–C
0x42 CH9–12 = D–G
Farm gate = CH0
Town gate = CH4
Yard servos = CH5–7

Each turnout has 3 Hall detectors
Common / Straight / Diverging

ROAD / SCENE MOTORS

Mine lift -> DRV8833 #1
Ski chairlift -> #2
Grain auger -> #3
Farm road conveyor -> #4
Ski lodge road conveyor -> #5
Long road conveyor -> #6
Town road conveyor -> #7
Rotary tipper -> #8

Each DRV8833 gets separate 1 A fast-blow 5 V branch

LOCOMOTIVE / IR SYSTEM

Each diesel loco:
12 V rail pickup -> 5 V buck
-> XIAO ESP32-C6
-> DRV8871 traction motor driver
-> 940 nm IR LED (~150 Ω)

Fixed IR checkpoints:
tunnel entrances/exits, tipper, yard stalls
Receiver input interface/channel map is NOT YET finalized
and is intentionally left unassigned here.

MASTER NOTES: one train per zone / one per destination; tipper fault/active blocks MAIN-3 and UND-1 except MAIN-2 -> Mine via A. Every railcar carries a magnetic strip; Hall detectors see each car.

*** New assignments added to complete this master pinout: GPIO12/13/16 yard relays and MCP 0x21 location inputs. These are software-remappable if you prefer other spare channels.**

HO SCALE RAILROAD — FULL PINOUT SCHEDULE

Controller revision: Raspberry Pi 4B + PCA9685 0x40/0x41/0x42 + MCP23017 0x20/0x21

RASPBERRY PI 4B — USED PINS

Physical	BCM / Rail	Assignment
Pin 1	3.3 V	Sensor rail for direct-to-Pi tipper sensors
Pin 2/4	5 V	Pi / logic supply as appropriate
Pin 3	GPIO2 / SDA1	I²C SDA -> bidirectional level shifter -> 5 V I²C bus
Pin 5	GPIO3 / SCL1	I²C SCL -> bidirectional level shifter -> 5 V I²C bus
Pin 29	GPIO5	Tipper Hall CLEAR, active-low
Pin 31	GPIO6	Tipper decoupler relay control
Pin 11	GPIO17	Tipper encoder pulse input
Pin 15	GPIO22	Tipper IR FRONT, clear = HIGH
Pin 16	GPIO23	Tipper IR BACK, clear = HIGH
Pin 18	GPIO24	Tipper Hall BACK, active-low
Pin 22	GPIO25	Tipper Hall FRONT, active-low
Pin 32	GPIO12	Yard Track 1 power relay control *
Pin 33	GPIO13	Yard Track 2 power relay control *
Pin 36	GPIO16	Yard Track 3 power relay control *
Pins 6/9/14/20/25/30/34/39	GND	Common/star ground bus

MCP23017 0x20 — INPUTS 1

0x20 Pin	Hall detector assignment
GPA0	Turnout A - COMMON
GPA1	Turnout A - STRAIGHT
GPA2	Turnout A - DIVERGING
GPA3	Turnout B - COMMON
GPA4	Turnout B - STRAIGHT
GPA5	Turnout B - DIVERGING
GPA6	Turnout C - COMMON
GPA7	Turnout C - STRAIGHT
GPB0	Turnout C - DIVERGING
GPB1	Turnout D - COMMON
GPB2	Turnout D - STRAIGHT
GPB3	Turnout D - DIVERGING
GPB4	Turnout E - COMMON
GPB5	Turnout E - STRAIGHT
GPB6	Turnout E - DIVERGING
GPB7	SPARE

POWER / WIRING RULES

- 12 V / 18 A track PSU: MAIN 7 A fast-blow; isolated TIPPER 2 A fast-blow.
- Tipper rails: 1/6 in air gap at both ends; own fused 12 V feed from same track PSU.
- 5 V / 18 A auxiliary PSU: each DRV8833 gets its own 1 A fast-blow branch; servo board gets 5 A branch.
- 16 V decoupler supply: fused, switched only through relay contacts; GPIO6 drives the relay interface.
- I²C uses GPIO2/GPIO3. Pi side is 3.3 V; use the bidirectional level shifter for the 5 V I²C peripheral bus.
- Common ground/star return. Keep high-current motor/track returns physically separate until the star point.

*** New master assignments used to close remaining gaps: yard relays = GPIO12/13/16; Mine/Farm/Ski/Town location Halls = MCP 0x21 GPA6/GPA7/GPB0/GPB1.**

PCA9685 0x40 — MOTOR M1

Channels	Motor	Load	Driver
CH0 / CH1	M1	Mine lift	DRV8833 #1
CH2 / CH3	M2	Ski chairlift	DRV8833 #2
CH4 / CH5	M3	Grain auger	DRV8833 #3
CH6 / CH7	M4	Farm road conveyor	DRV8833 #4
CH8 / CH9	M5	Ski lodge road conveyor	DRV8833 #5
CH10 / CH11	M6	Long-haul road conveyor	DRV8833 #6
CH12 / CH13	M7	Town road conveyor	DRV8833 #7
CH14 / CH15	M8	Rotary tipper motor	DRV8833 #8

PCA9685 0x42 — SERVOS

Channel	Servo / Actuator
CH0	Farm gate
CH1	Turnout A
CH2	Turnout B
CH3	Turnout C
CH4	Town gate
CH5	Yard 1
CH6	Yard 2
CH7	Yard 3
CH8	Tipper wheel-lock servo
CH9	Turnout D
CH10	Turnout E
CH11	Turnout F
CH12	Turnout G
CH13	SPARE
CH14	SPARE
CH15	SPARE

PCA9685 0x41 — MOTOR M2 / EXPANSION

CH0–CH15 currently unassigned / spare. Keep address 0x41 reserved for future motor outputs.

Do not duplicate any 0x40 motor assignment on 0x41.

MCP23017 0x21 — INPUTS 2

0x21 Pin	Hall detector assignment
GPA0	Turnout F - COMMON
GPA1	Turnout F - STRAIGHT
GPA2	Turnout F - DIVERGING
GPA3	Turnout G - COMMON
GPA4	Turnout G - STRAIGHT
GPA5	Turnout G - DIVERGING
GPA6	MINE location Hall *
GPA7	FARM location Hall *
GPB0	SKI location Hall *
GPB1	TOWN location Hall *
GPB2	SPARE
GPB3	SPARE
GPB4	SPARE
GPB5	SPARE
GPB6	SPARE